

Urban Air Mobility MAS Project Report

1. Agent Roles and Behaviors

1.1 CustomerAgent (JourneyRequester)

- **Role:** Submits journey requests and receives route proposals.
- **Behaviors:**
 - **OneShotBehaviour:** Sends REQUEST_JOURNEY:X,Y,30min to the Supervisor agent.
 - **CyclicBehaviour:** Listens for PROPOSE_ROUTE messages and logs received proposals.

1.2 SupervisorSoSAgent (JourneyCoordinator)

- **Role:** Coordinates journey planning and negotiates resource dispatch.
- **Behaviors:**
 - **CyclicBehaviour:** Waits for REQUEST_JOURNEY from Customer; sends DISPATCH_REQUEST to Scooter.
 - **CyclicBehaviour:** Processes Scooter's ACCEPT or REJECT, then sends PROPOSE_ROUTE back to Customer.

1.3 ScooterAgent (Vehicle Resource)

- **Role:** Decides whether to accept or reject dispatch requests based on internal state.
- **Behaviors:**
 - **CyclicBehaviour:** Waits for DISPATCH_REQUEST; if not busy, sets busy and replies ACCEPT, otherwise REJECT.

2. Communication Flow

1. **CustomerAgent** → SupervisorSoSAgent: REQUEST_JOURNEY:X,Y,30min
2. **SupervisorSoSAgent** → ScooterAgent: DISPATCH_REQUEST
3. **ScooterAgent** → SupervisorSoSAgent: ACCEPT or REJECT
4. **SupervisorSoSAgent** → CustomerAgent: PROPOSE_ROUTE:scooter1
ETA=10min
5. **CustomerAgent** logs the proposal.